



CWIT R23

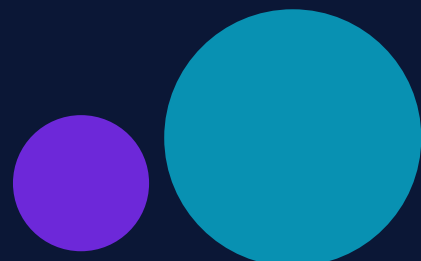
Semester 3 to 6 YouTube Lecture Guide

Curated tutorial and lecture resources for **Computer Engineering (CP)** and **Computer Engineering & IoT (CI)** at Cusrow Wadia Institute of Technology.

COVERS	36 subject/elective groups across Semesters 3-6
LINKS	72 clickable YouTube lectures, courses and playlists
CHECKED	28 June 2026

Prepared for Lernio resource integration

Public YouTube resources; not an official endorsement by CWIT or MSBTE.



How to use this guide

Start with the **Primary** resource. Use an alternate when you need another explanation, more examples or a different teaching style. For programming, networking, databases, microcontrollers, IoT and testing, reproduce each practical step yourself instead of only watching.

PRIMARY	Recommended starting point for the subject.
ALTERNATE	Second explanation, deeper theory or extra practical work.
ELECTIVE	Study only the elective officially allotted to your class.
PROJECT	Support material; the CWIT rubric and faculty guide remain authoritative.

Subject index

3.1	Data Structures	Semester 3
3.2	Object Oriented Programming with C++	Semester 3
3.3	User Interface Programming	Semester 3
3.4	Animation Techniques	Semester 3
3.5	Microprocessors and Its Programming - CP Only	Semester 3
3.6	Data Communication - CP Only	Semester 3
3.7	Digital Techniques and Microcontroller - CI Only	Semester 3
3.8	IoT and Its Application - CI Only	Semester 3
4.1	Operating System - CP	Semester 4
4.2	Embedded Operating Systems - CI	Semester 4
4.3	Database Management System	Semester 4
4.4	Computer Networks / Computer Networks and Security	Semester 4
4.5	Python Programming	Semester 4
4.6	Java Programming	Semester 4
4.7	IoT Architecture and Protocols - CI Only	Semester 4
4.8	Entrepreneurship Development and Marketing Management	Semester 4
5.1	Software Engineering	Semester 5
5.2	Advanced Java Programming	Semester 5
5.3	PHP Programming - CP Only	Semester 5
5.4	Recent Trends in Computer Networks - CP Elective	Semester 5
5.5	Database Architecture and Emerging Technologies - CP Elective	Semester 5
5.6	Fundamentals of Data Science - CP Elective	Semester 5
5.7	Wireless Ad Hoc Network - CI	Semester 5
5.8	IoT in Robotics - CI	Semester 5

5.9	Seminar, Capstone Initiation and Internship Support	Semester 5
5.10	CI Elective Group A - Cyber Security / Cloud Computing / Data Analytics	Semester 5
6.1	Industrial Management for IT Industry	Semester 6
6.2	Emerging Trends in Computer Engineering - CP	Semester 6
6.3	Machine Learning - CP	Semester 6
6.4	AI Tools and Techniques	Semester 6
6.5	Software Testing Tools and Automation	Semester 6
6.6	Network and Information Security - CP Elective	Semester 6
6.7	Cloud Computing - CP/CI Elective	Semester 6
6.8	Data Analytics and Visualization using R - CP Elective	Semester 6
6.9	CI Elective Group B - Information Security / Cloud Security / Big Data Analytics	Semester 6
6.10	Capstone Project Support	Semester 6

Important branch/elective note: CP and CI share several core subjects but also have different hardware, IoT and elective courses. Exact elective allotment and the placement of some CI advanced courses should always be checked against the latest departmental timetable.

SEMESTER 3

Core programming, data structures, interfaces and CP/CI branch foundations

Data Structures

CRITICAL

R23CP2402 / R23CI2602 - Shared

Coverage focus: Arrays, linked lists, stacks, queues, trees, searching, sorting, traversal and time-complexity basics.

PRIMARY - EXAM ORIENTED

Data Structure - Complete Course

Gate Smashers | Language: Hindi / Hinglish

Short, structured explanations with algorithm tracing and exam-focused examples.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLxCzCOWd7aiEwaANNt3OqJPVlxwp2ebiT>

ALTERNATE - DEEPER

Data Structures and Algorithms

Jenny's Lectures CS/IT | Language: English / Hinglish

Useful for linked lists, stacks, queues, trees and step-by-step program logic.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLhXQeMKeH2-KpEXsbYzFFAeN6acs5ifAG>

Object Oriented Programming with C++

CRITICAL

R23CP6404 / R23CI6604 - Shared

Coverage focus: Classes, objects, constructors, inheritance, polymorphism, operator overloading, templates and file handling.

PRIMARY

C++ Programming

Neso Academy | Language: English

Clear chapter-wise coverage from C++ basics through object-oriented concepts.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLBlnK6fEyqRh6isJ01MBnbNpV3ZsktSyS>

ALTERNATE - PRACTICE

C++ Placement Course

Jenny's Lectures CS/IT | Language: English / Hinglish

Adds coding practice and another explanation of OOP concepts.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLdo5W4Nhv31YU5Wx1dopka58teWP9aCee>

User Interface Programming

PRACTICAL

R23CP2405 / R23CI2605 - Shared

Coverage focus: GUI components, layouts, events, forms, menus and creation of small desktop applications.

PRIMARY - JAVA GUI

Java Swing GUI - Full Course

Bro Code | Language: English

Practical screen-based course for windows, buttons, panels, layouts and event handling.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/watch?v=Kmgo00avvEw>

ALTERNATE - PYTHON GUI

Python Tkinter GUI Design - Complete Course

YouTube full course | Language: English

Useful where the laboratory uses Tkinter or for understanding common GUI patterns.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/watch?v=0tM-I_ZsxjU

Important: Confirm the GUI language/framework used by your teacher before choosing the Java Swing or Tkinter track.

Animation Techniques

PRACTICAL

R23CP1402 / R23CI1602 - Shared

Coverage focus: Animation principles, keyframes, timing, motion, 2D assets and practical animation workflow.

PRIMARY - 2D

Learn 2D Animation in Blender

YouTube playlist | Language: English

Practical beginner path for keyframes, drawing and 2D animation workflow.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLgGbWld6zgaWNZu4bG-DvAWsN1G1GYUiw>

ALTERNATE

Blender 2D Grease Pencil Tutorials

YouTube playlist | Language: English

More focused tutorials for Grease Pencil tools and short animation exercises.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLeaNmybb7X-vUsE24RyEBGqhfAlx4LxXz>

Microprocessors and Its Programming - CP Only

HIGH PRIORITY

R23CP2403

Coverage focus: Processor architecture, registers, memory, addressing modes, instruction set and assembly-language programming.

PRIMARY

Microprocessors and Microcontrollers

Neso Academy | Language: English

Strong conceptual coverage of architecture, instruction sets and interfacing.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLBlnK6fEyqRgyFCCgqdcBowmSp_BTks4F

ALTERNATE - HINGLISH

Microprocessor and Microcontroller

Gate Smashers | Language: Hindi / Hinglish

Exam-friendly explanations and another route through architecture and programming topics.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLBIfK3DghyO6SVpIKk3QnToRTawKj6LaF>

Data Communication - CP Only

IMPORTANT

R23CP2404

Coverage focus: Signals, transmission media, encoding, multiplexing, switching, error detection/correction and data-link concepts.

PRIMARY

Data Communication Networks

YouTube playlist | Language: English

Focused coverage of data-communication principles and common transmission topics.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PL-6c20OcmtA3xDpzoHqOHOIVT01UFRFI>

ALTERNATE

Computer Networks

Neso Academy | Language: English

Use the physical and data-link layer lectures for deeper explanations and numericals.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLBlnK6fEyqRgMCUAG0XRw78UA8qnv6jEx>

Digital Techniques and Microcontroller - CI Only

CRITICAL FOR IOT

R23CI2603

Coverage focus: Number systems, Boolean algebra, combinational/sequential circuits, registers, counters and microcontroller fundamentals.

PRIMARY - DIGITAL

Digital Electronics

Neso Academy | Language: English

Clear foundation for gates, Boolean algebra, combinational and sequential logic.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLBlnK6fEyqRjYKHlzipvuuZMTIJKsvQK>

MICROCONTROLLER ALTERNATE

Microprocessors and Microcontrollers

Neso Academy | Language: English

Use the relevant microcontroller architecture and programming lectures after digital fundamentals.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLBlnK6fEyqRgyFCCgqdcBowmSp_BTks4F

IoT and Its Application - CI Only

CRITICAL FOR IOT

R23CI2604

Coverage focus: IoT building blocks, sensors, connectivity, cloud, MQTT, applications and introductory hardware projects.

PRIMARY - SYLLABUS STYLE

Internet of Things - Complete Playlist

University-oriented IoT playlist | Language: English / Hinglish

Covers IoT concepts, architecture, devices, protocols and applications in a semester-friendly order.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLdguAlzSoZprsOmvd_ZeuUM-5HeVFZlke

PRACTICAL ALTERNATE

Learn Robotics and IoT Tutorials

Learn Robotics | Language: English

Adds practical Arduino, Node-RED, MQTT and sensor-oriented demonstrations.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLYxfF2qa9dis7Db06-a2RNFhOaUTBScWn>

SEMESTER 4

Operating systems, databases, networks, Python/Java and IoT architecture

Operating System - CP

CRITICAL

R23CP2406

Coverage focus: Processes, CPU scheduling, synchronization, deadlocks, memory management, file systems and I/O.

PRIMARY - EXAM ORIENTED

Operating System - Complete Playlist

Gate Smashers | Language: Hindi / Hinglish

Comprehensive semester-exam coverage with numericals and clear diagrams.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLxCzCOWd7aiGz9donHRrE9l3Mwn6XdP8p>

ALTERNATE - FULL COURSE

Operating Systems Course for Beginners

freeCodeCamp.org | Language: English

Long-form conceptual course for deeper understanding beyond exam revision.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/watch?v=yK1uBHPdp30>

Embedded Operating Systems - CI

CRITICAL FOR IOT

R23CI2606

Coverage focus: Real-time systems, tasks, scheduling, queues, semaphores, mutexes, interrupts and FreeRTOS basics.

PRIMARY - RTOS

RTOS Fundamentals and FreeRTOS

Embedded systems playlist | Language: English

Introduces real-time concepts and practical FreeRTOS configuration and task management.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLmdxycs9ZImdo9VA_Cg7rbJpWLTnCYcZ2

PRACTICAL ALTERNATE

FreeRTOS for Microcontrollers

Learn Embedded Systems | Language: English

Hands-on task, queue, semaphore and mutex examples for microcontroller boards.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PL8IID45FIDTQz6Du4nUfaqJvxOwz86xik>

Database Management System

CRITICAL

R23CP2407 / R23CI2607 - Shared

Coverage focus: ER model, relational model, SQL, normalization, transactions, concurrency, indexing and recovery.

PRIMARY - EXAM ORIENTED

Database Management System - Complete Playlist

Gate Smashers | Language: Hindi / Hinglish

Strong theory, normalization, transactions and semester-exam preparation.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLxCzCOWd7aiFAN6I8CuViBuCdJgiOkT2Y>

ALTERNATE - CHAPTER WISE

Database Management Systems

Neso Academy | Language: English

Clear topic-by-topic explanations that work well for revision and difficult concepts.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLBlnK6fEyqRiYryTrbKHX1Sh9luYI0dhX>

Computer Networks / Computer Networks and Security

CRITICAL

R23CP2408 / R23CI2608

Coverage focus: OSI/TCP-IP, Ethernet, addressing, subnetting, routing, transport/application protocols and introductory security.

PRIMARY

Computer Networks

Neso Academy | Language: English

Detailed layer-by-layer coverage with clear diagrams and protocol explanations.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLBlnK6fEyqRgMCUAG0XRw78UA8qnv6jEx>

SECURITY EXTENSION

Network Security and Cryptography

Gate Smashers | Language: Hindi / Hinglish

Use for cryptography, keys, attacks and security topics in the CI course.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLxCzCOWd7aiFjkQbGj_yHeTEVPn422_z5

Python Programming

HIGH PRIORITY

R23CP1403 / R23CI1603 - Shared

Coverage focus: Python syntax, control flow, collections, functions, modules, files, exceptions and introductory OOP.

PRIMARY - HINDI

Python Tutorials for Absolute Beginners

CodeWithHarry | Language: Hindi / Hinglish

Beginner-friendly coding demonstrations and a practical progression through Python.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLu0W_9lII9agIcNt8t4iYVSZ3eykIAOME

ALTERNATE - ENGLISH

Data Science and Python Foundations

freeCodeCamp.org | Language: English

Use the Python basics and data-handling videos for additional practice.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLWKjhJtqVAblQe2CCWqV4Zy3LY01Z8aF1>

Java Programming

HIGH PRIORITY

R23CP1404 / R23CI1604 - Shared

Coverage focus: Java syntax, classes, objects, inheritance, interfaces, packages, exceptions, collections and basic GUI/database work.

PRIMARY

Java Tutorial for Beginners

Telusko | Language: English / Indian English

Clear step-by-step Java programming course with practical coding demonstrations.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLsyeobzWxl7pe_liTfNyr55kwJPWbgxB5

ALTERNATE

Java Programming Tutorial

Telusko | Language: English / Indian English

Additional structured lectures and practice examples for core Java concepts.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLsyeobzWxl7pFZoGT1NbZJpywedeyzyaf>

IoT Architecture and Protocols - CI Only

CRITICAL FOR IOT

R23CI2609

Coverage focus: Layered IoT architecture, MQTT, CoAP, HTTP, device-to-cloud communication, gateways and end-to-end systems.

PRIMARY

IoT End-to-End Architecture

IoT architecture playlist | Language: English

Architecture-focused series covering how devices, gateways, networks and cloud services connect.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLCCWsGv21PgVDS3S1InRbmWCqudrZlBU>

ALTERNATE - BROADER

Internet of Things - Complete Playlist

University-oriented IoT playlist | Language: English / Hinglish

Use protocol and architecture lectures to reinforce MQTT, connectivity and IoT models.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLdguAlzSoZprsOmvd_ZeuUM-5HeVFZlke

Entrepreneurship Development and Marketing Management

SKILL SUPPORT

R23CP6405 / R23CI6605 - Shared

Coverage focus: Entrepreneurship, business ideas, market research, project planning, finance basics, marketing and small-enterprise management.

PRIMARY - ENGINEERING MANAGEMENT

Engineering Management and Economics

University lecture playlist | Language: English

Supports planning, organization, economics and management concepts used in entrepreneurship subjects.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLXPm4k-8ERi046Ltkl1hcbacCeBTNbdf>

PROJECT PLANNING ALTERNATE

Software Project Management - Complete Course

Exam and industry course | Language: Hindi / Hinglish

Useful for project planning, estimation, scheduling, risk and team management.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PL4S7L6ygAUXfdVskyR5gMT5jC8o9VTx49>

SEMESTER 5

Software engineering, advanced development, specialization and project initiation

Software Engineering

CRITICAL

R23CP2409 / R23CI2610 - Shared

Coverage focus: SDLC models, requirements, analysis/design models, testing, estimation, risk, quality and maintenance.

PRIMARY - EXAM ORIENTED

Software Engineering - Complete Playlist

Gate Smashers | Language: Hindi / Hinglish

Complete semester-oriented coverage with models, diagrams and exam-relevant explanations.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLxCzCOWd7aiEed7SKZBnC6ypFDWYLRvB2>

ALTERNATE - PROJECT MANAGEMENT

Software Engineering and Project Management

YouTube playlist | Language: Hindi / Hinglish

Adds project planning, estimation and management topics around the software lifecycle.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLYwpaL_SFmcCB7zUM0YSDR-1mM4KoilyLM

Advanced Java Programming

HIGH PRIORITY

R23CP1405 / R23CI2613

Coverage focus: JDBC, Swing/AWT, networking, servlets, JSP and building database-backed Java applications.

PRIMARY

Advanced Java - Servlet, JSP and JDBC

YouTube course playlist | Language: English / Hinglish

Practical progression through JDBC, web components and database-connected Java applications.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLR77UJiTj7USmpKgZSSGwc7WjLZfm5xN>

ALTERNATE

Advanced Java Programming

YouTube playlist | Language: English / Hinglish

Additional coverage for Swing, JDBC, servlets and related advanced Java topics.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLrW5vJdpAaLr8RANKMTjwayB7dc4pO46H>

Important: The exact semester placement of R23CI2613 should be checked against the current CI timetable; the course is included here with the advanced-semester resources.

PHP Programming - CP Only

HIGH PRIORITY

R23CP1406

Coverage focus: PHP syntax, forms, sessions, cookies, database connectivity, CRUD and simple web applications.

PRIMARY

PHP and MySQL - Full Stack Course

YouTube playlist | Language: English / Hinglish

Covers PHP from basics through MySQL integration and CRUD application work.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLZ8-rdXtysy5zLdQ6uX7VDQFF5koiGLe7>

ALTERNATE - HINDI

PHP Course and Web Development Videos

CodeWithHarry playlist | Language: Hindi / Hinglish

Use the PHP course videos for an alternate practical explanation.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLu0W_9lII9ahKZ42vg2w9ERPmShYbYAB7

Recent Trends in Computer Networks - CP Elective

ELECTIVE

R23CP3401

Coverage focus: Network virtualization, SDN, data-center networking, IoT networking and emerging network technologies.

PRIMARY - SDN

Software Defined Networking

Nick Feamster lecture series | Language: English

Detailed academic course on SDN, control/data planes and OpenFlow.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLpherdrLyny8YN4M24iRJBMCXkLcGbmhY>

ALTERNATE - RECENT NETWORKING

Recent Advances in Networking: SDN, Big Data and IoT

Academic lecture series | Language: English

Broader treatment of data-center networking, virtualization, SDN and IoT trends.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLjGG94etKypLHyBN8mOgwJLHD2FFIMGq5>

Database Architecture and Emerging Technologies - CP Elective

ELECTIVE

R23CP3402

Coverage focus: Advanced database architecture, distributed/NoSQL systems, MongoDB, scalability and emerging database models.

PRIMARY - NOSQL

MongoDB - NoSQL Database Tutorial

YouTube playlist | Language: English

Beginner-friendly MongoDB course covering document databases, queries and practical database work.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLLAZ4kZ9dFpOFJ9JcVW9u4PISWO-VFoao>

ALTERNATE - HINDI

MongoDB and Mongoose Full Course

YouTube playlist | Language: Hindi / Hinglish

Practical MongoDB CRUD, queries and application integration.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PL2PkZdv6p7ZnPIMuAVv3QckkrpgECa6d3>

Fundamentals of Data Science - CP Elective

ELECTIVE

R23CP3403

Coverage focus: Data-science workflow, Python tools, data cleaning, visualization, statistics and introductory machine learning.

PRIMARY

Data Science

freeCodeCamp.org | Language: English

Collection of substantial courses on Python, data analysis, visualization, statistics and machine learning.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLWKjhJtqVAblQe2CCWqV4Zy3LY01Z8aF1>

ALTERNATE - HINDI

Python for Data Science

CodeWithHarry | Language: Hindi / Hinglish

Practical Python data-science foundation with common libraries and examples.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLu0W_9lI9agK8pojo23OHnZ3Jm6VQCH

Wireless Ad Hoc Network - CI

HIGH PRIORITY

R23CI2611

Coverage focus: MANET concepts, wireless MAC, routing protocols, mobility, topology changes, performance and simulation.

PRIMARY

Mobile Computing and MANET

YouTube lecture playlist | Language: English / Hinglish

Includes mobile ad-hoc networks, MAC concepts and routing protocols.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLY3TW4ACyv_YqRF54UuACh4h0o98p8Ljd

PRACTICAL ALTERNATE

NS-3 Tutorials - MANET and Wireless Networks

Engineering Clinic | Language: English

Useful for simulation-oriented practicals and routing-protocol demonstrations.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLX6MKaDw0naZILVYjo8_quA4Jl8Jz-idL

IoT in Robotics - CI

PRACTICAL

R23CI2612

Coverage focus: Sensors, motors, controllers, Arduino/Raspberry Pi, remote control, MQTT and connected robotic projects.

PRIMARY

Arduino Projects and Tutorials

Free Arduino courses | Language: English

Beginner-to-practical Arduino and robotics projects for sensors, motors and control.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PL4g1oAdmuCfrl88GPnJHCJNi0vr416XmZ>

ALTERNATE - PROJECTS

Arduino, Raspberry Pi and Robotics Projects

Project playlist | Language: English

Connected robot, Raspberry Pi and embedded-project demonstrations for deeper practice.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PL0yJV-YxkLaUNxdPCtPE_IcEcNz5DLDxw

Seminar, Capstone Initiation and Internship Support

PROJECT SUPPORT

R23CP1407, R23CP5401 / R23CI1605 and CI internship-project requirements

Coverage focus: Selecting a problem, planning scope, documenting work, version control, presentation and project tracking.

PRIMARY - PLANNING

Software Project Management - Full Course

YouTube playlist | Language: Hindi / Hinglish

Useful for scope, scheduling, risk, estimation, documentation and team coordination.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PL2zqvGhQWd_NFuos0C0E0ILhvuT6u13IC

PRACTICAL - VERSION CONTROL

Git and GitHub Tutorial

Curated beginner playlist | Language: Hindi / English

Use for storing project code, collaboration, commits, branches and submission history.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PL7T88QvMOZMhxnHI5KJuw3uUakiXvrgiJ>

Important: Seminar, internship and capstone marks depend on your institute rubric, guide approval, documentation and actual project work; videos are only support material.

CI Elective Group A - Cyber Security / Cloud Computing / Data Analytics

ELECTIVE

R23CI3601 / R23CI3602 / R23CI3603

Coverage focus: Choose only the playlist matching the elective allotted to your class.

CYBER SECURITY

Cyber Security - Updated Playlist

Simplilearn | Language: English

Foundation in threats, attacks, firewalls, phishing, ransomware and security practices.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLEiEAq2VkUUJfPOj5nRounXvf3n17PCft>

DATA ANALYTICS

Power BI Beginner to Advanced - Hindi

Industries Helping Hands | Language: Hindi / Hinglish

Practical data import, modelling, DAX, visualizations and dashboard work.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLxzTa0VPR9rw4QEQ49PdsFd3SmVXDR3-i>

Important: For Cloud Computing, also use the dedicated cloud playlist listed under Semester 6 electives in this guide.

SEMESTER 6

Machine learning, AI, testing, management, security/cloud/data electives and capstone

Industrial Management for IT Industry

IMPORTANT

R23CP1408 / R23CI1606 - Shared

Coverage focus: Organization, leadership, operations, quality, inventory, project management, HR and IT-industry practices.

PRIMARY

Engineering Management and Economics

University lecture playlist | Language: English

Structured management foundation relevant to engineering and IT organizations.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLXPm4k-8ERi046Ltkll1hcbacCeBTNbdf>

ALTERNATE - PROJECT MANAGEMENT

Software Project Management Hindi/Urdu

Software Engineering channel | Language: Hindi / Urdu

Useful for planning, teams, risk, estimation and project-control topics.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLPtLP7KjDoqVRWHPQI1stwbKbr-OCG3I>

Emerging Trends in Computer Engineering - CP

IMPORTANT

R23CP2410

Coverage focus: AI/ML, cloud, IoT, blockchain, edge computing, modern software platforms and current computing trends.

PRIMARY - AI FOUNDATION

Artificial Intelligence

Gate Smashers | Language: Hindi / Hinglish

Semester-friendly introduction to AI concepts, search, agents and neural networks.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLxK1FVS_3IoL3TL_IzI6mFs4kJjENqdtg

NETWORKING TRENDS

Recent Advances in Networking: SDN, Big Data and IoT

Academic lecture series | Language: English

Useful for SDN, virtualization, IoT and data-center networking trends.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLjGG94etKypLHyBN8mOgwJLHD2FFIMGq5>

Important: Emerging-trends syllabi change quickly. Compare the current unit list with these resources before studying every video.

Machine Learning - CP

CRITICAL

R23CP2411

Coverage focus: Data preparation, supervised/unsupervised learning, regression, classification, clustering, evaluation and Python implementation.

PRIMARY - HINGLISH

Machine Learning

Gate Smashers | Language: Hindi / Hinglish

Compact semester-style sequence covering core ML algorithms and concepts.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLnXlbeknlQabSODsSrViSCRYdNzII3S7G>

ALTERNATE - FULL COURSE

Complete Machine Learning Course

Siddhardhan | Language: English / Indian English

Long-form practical Python course with implementation and projects.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/watch?v=LcWFedjaR4Q>

AI Tools and Techniques

HIGH PRIORITY

R23CP6406 / R23CI6606 - Shared

Coverage focus: AI concepts, practical tools, prompt-based workflows, data/ML utilities, responsible use and small AI applications.

PRIMARY - AI THEORY

Artificial Intelligence

Gate Smashers | Language: Hindi / Hinglish

Builds the conceptual foundation needed before using AI tools mechanically.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLxK1FVS_3IoL3TL_IzI6mFs4kJjENqdtg

ALTERNATE - DEEP LEARNING

Deep Learning - ANN, RNN and Transformers

Gate Smashers | Language: Hindi / Hinglish

Use selected videos for neural-network and modern AI technique topics.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLxCzCOWd7aiHQ9Qyo70JCTkTxczPnVC6s>

Important: Use only institution-approved AI tools for assessed work, and keep a record of prompts, sources and your own modifications.

Software Testing Tools and Automation

CRITICAL

R23CP6407 / R23CI6607 - Shared

Coverage focus: Testing concepts, test cases, defect lifecycle, unit/integration/system testing, Selenium and test automation frameworks.

PRIMARY - SELENIUM JAVA

Learn Selenium Automation with Java - Full Course

Automation Step by Step | Language: English / Indian English

Complete practical Selenium WebDriver progression from basics to framework concepts.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLuAoMvvRllpSIX-RmzvABaB-ct1Iq75Br>

ALTERNATE - TESTING FOUNDATION

Software Testing Training Videos

Eureka | Language: English

Covers manual testing, automation basics and Selenium-oriented concepts.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PL9ooVrP1hQOHYamtWSSIt5jF52WvP0cHI>

Network and Information Security - CP Elective

ELECTIVE

R23CP3404

Coverage focus: Security goals, attacks, cryptography, authentication, network defence, firewalls and secure communication.

PRIMARY - EXAM ORIENTED

Network Security and Cryptography

Gate Smashers | Language: Hindi / Hinglish

Concise, exam-friendly cryptography and network-security explanations.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLxCzCOWd7aiFjkQbGj_yHeTEVPn422_z5

ALTERNATE - BROADER

Cryptography and Network Security

YouTube course playlist | Language: English

Adds attacks, goals, cryptographic components and security mechanisms.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLiineyru3vMCz3AExYPOFmR_DrZploOOE

Cloud Computing - CP/CI Elective

ELECTIVE

R23CP3405 / R23CI3602

Coverage focus: Cloud characteristics, service/deployment models, virtualization, architecture, storage, security and common platforms.

PRIMARY - HINGLISH

Cloud Computing - Gate Smashers Course

Gate Smashers channel playlist area | Language: Hindi / Hinglish

Start from the linked cloud lecture and continue through the numbered cloud-computing series.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/watch?v=aaL61HouKLs>

ALTERNATE - FULL COURSE

Cloud Computing Full Course

Simplilearn | Language: English

Broad end-to-end cloud foundation and current platform concepts.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/watch?v=q3m1AB9ECXo>

Data Analytics and Visualization using R - CP Elective

ELECTIVE

R23CP3406

Coverage focus: R/RStudio, data frames, cleaning, descriptive analysis, charts, ggplot-style visualization and reporting.

PRIMARY - R

R Programming for Data Analytics

YouTube playlist | Language: English

RStudio setup, packages, data handling and analysis workflow for beginners.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLnGaO7jfhHOvalh3Jt48Ni4IsRcHBGbHHn>

ALTERNATE - DATA SCIENCE

Data Science

freeCodeCamp.org | Language: English

Includes R basics and broader data-visualization courses for additional depth.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLWKjhJtqVAblQe2CCWqV4Zy3LY01Z8aF1>

CI Elective Group B - Information Security / Cloud Security / Big Data Analytics

ELECTIVE

R23CI3604 / R23CI3605 / R23CI3606

Coverage focus: Choose only the resource matching your allotted elective.

CLOUD SECURITY

Free Cloud Security Training

Cloud-security training playlist | Language: English

Security controls, cloud risks, identity, architecture and defensive practices.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PLOWdy-NBQHJsGVJJ0OpBpxTbPW1v99Y2a>

BIG DATA ANALYTICS

Big Data Hadoop - Full Course

YouTube course playlist | Language: English

Introduces big-data concepts and the Hadoop ecosystem in a compact course.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PL6UwySlcwEYJ2hFuGlv4VEHUAfl-GCNT>

Important: For Information Security, use the Network and Information Security playlists directly above.

Capstone Project Support

PROJECT SUPPORT

R23CP5402 / CI capstone requirement - verify current CI code

Coverage focus: Requirements, architecture, implementation, Git workflow, testing, report writing, demo preparation and final presentation.

PRIMARY - PROJECT MANAGEMENT

Software Engineering with Project Management

Ekeeda | Language: English / Hinglish

Connects software-engineering concepts with planning, estimation, tracking and delivery.

OPEN YOUTUBE RESOURCE

https://www.youtube.com/playlist?list=PLm_MSCLsnwm8jU0kPRug2n7Yv4oSk7HLk

PRACTICAL - GIT/GITHUB

Git and GitHub Tutorial

Curated beginner playlist | Language: Hindi / English

Essential for version control, collaboration, backup and a clean project history.

OPEN YOUTUBE RESOURCE

<https://www.youtube.com/playlist?list=PL7T88QvMOZMhxnHI5KJuw3uUakiXrvgiJ>

Important: Your capstone must follow the latest CWIT project rubric and your guide's approved scope. Do not copy a tutorial project unchanged.

Verification and limitations

- 1 The CP Semester 3-6 subject mapping follows the official CWIT R23 Computer Engineering curriculum. The CI mapping uses the official R23 Computer Engineering and IoT curriculum together with published examination scheduling information.
- 2 YouTube links were selected and checked through web search on 28 June 2026. Titles, playlist order, availability, advertisements and individual videos can change after that date.
- 3 A playlist may teach the correct subject in a different unit order. Compare it with the latest CWIT unit syllabus, class notes, laboratory manual and teacher instructions.
- 4 Elective groups are not compulsory as a whole. Study only the elective allotted to your batch and semester.
- 5 Value-education subjects such as Indian Constitution and ethics are not given dedicated technical playlists here because internal activities and institute notes are normally more important for assessment.
- 6 Project, internship, seminar and capstone resources are guidance only. Original implementation, documentation, testing, attendance, reviews and guide approval determine assessment.

Official curriculum references

[CWIT R23 - Computer Engineering curriculum](https://cwit.mespune.org/wp-content/uploads/2021/07/COMPUTER-MPECS-23-CURRICULUM.pdf)

<https://cwit.mespune.org/wp-content/uploads/2021/07/COMPUTER-MPECS-23-CURRICULUM.pdf>

[CWIT R23 - Computer Engineering and IoT curriculum](https://cwit.mespune.org/wp-content/uploads/2023/07/IOTR23_ALL_Curriculum-FINALV.pdf)

https://cwit.mespune.org/wp-content/uploads/2023/07/IOTR23_ALL_Curriculum-FINALV.pdf

[CWIT Winter Examination 2025 timetable](https://cwit.mespune.org/wp-content/uploads/2025/10/Winter-Examination-2025.pdf)

<https://cwit.mespune.org/wp-content/uploads/2025/10/Winter-Examination-2025.pdf>

End of guide

Every resource link is both clickable and printed in full so it can be copied from the PDF.